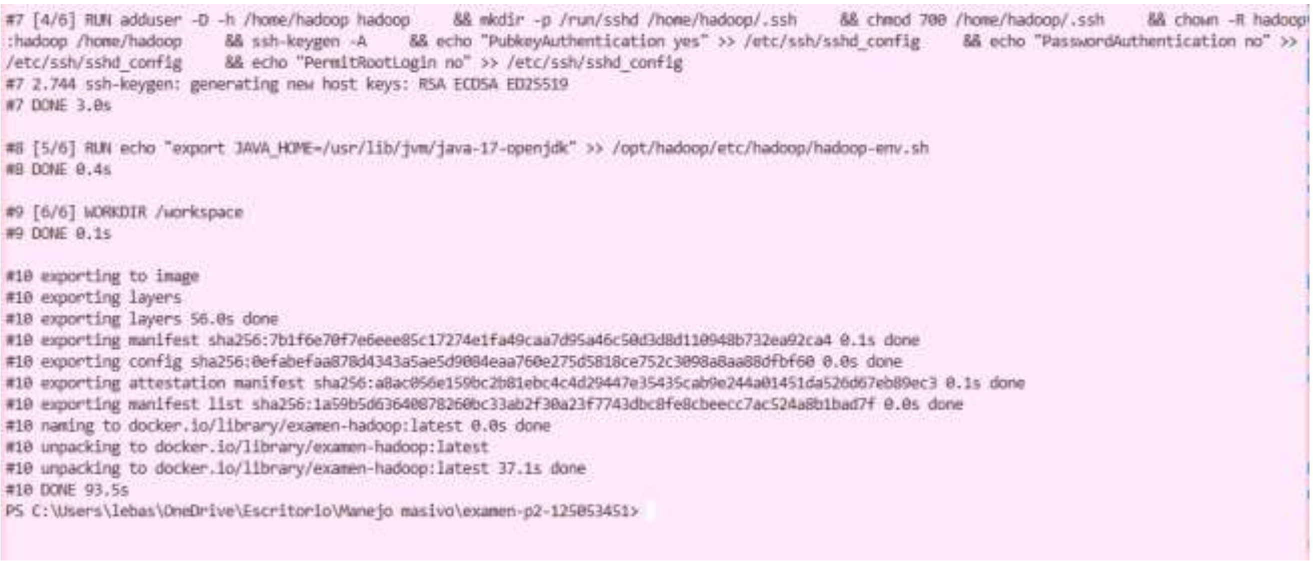
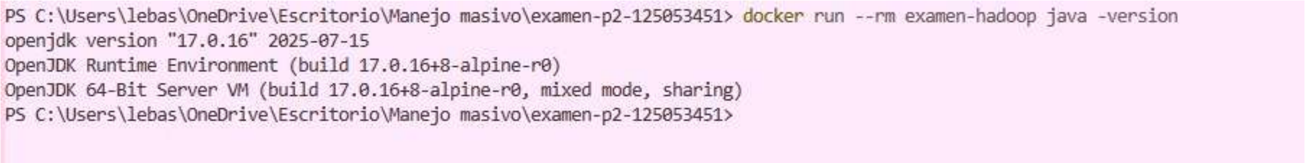
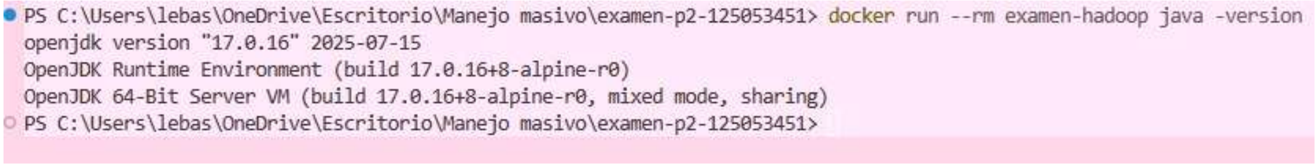
Evidencias, examen.



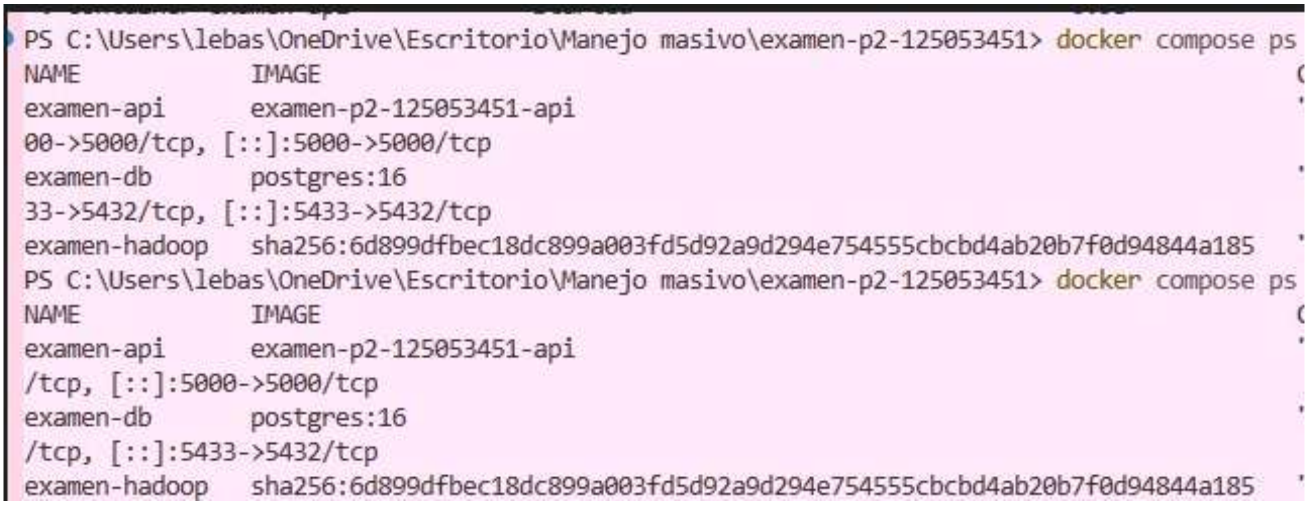
```
#7 [4/6] RUN adduser -D -h /home/hadoop hadoop     && mkdir -p /run/sshd /home/hadoop/.ssh     && chmod 700 /home/hadoop/.ssh     && chown -R hadoop
:hadoop /home/hadoop     && ssh-keygen -A     && echo "PubkeyAuthentication yes" >> /etc/ssh/sshd_config     && echo "PasswordAuthentication no" >>
/etc/ssh/sshd_config     && echo "PermitRootLogin no" >> /etc/ssh/sshd_config
#7 2.744 ssh-keygen: generating new host keys: RSA ECDSA ED25519
#7 DONE 3.0s

#8 [5/6] RUN echo "export JAVA_HOME=/usr/lib/jvm/java-17-openjdk" >> /opt/hadoop/etc/hadoop/hadoop-env.sh
#8 DONE 0.4s

#9 [6/6] WORKDIR /workspace
#9 DONE 0.1s

#10 exporting to image
#10 exporting layers
#10 exporting layers 56.0s done
#10 exporting manifest sha256:7b1f6e70f7e6eee85c17274e1fa49caa7d95a46c50d3d8d110948b732ea92ca4 0.1s done
#10 exporting config sha256:0efabefaa878d4343a5ae5d9084eaa760e275d5818ce752c3098a8aa88dfbf60 0.0s done
#10 exporting attestation manifest sha256:a8ac056e159bc2b81ebc4c4d29447e35435cab9e244a01451da526d67eb89ec3 0.1s done
#10 exporting manifest list sha256:1a59b5d63640878260bc33ab2f30a23f7743dbc8fe8cbeecc7ac524a8b1bad7f 0.0s done
#10 naming to docker.io/library/examen-hadoop:latest 0.0s done
#10 unpacking to docker.io/library/examen-hadoop:latest
#10 unpacking to docker.io/library/examen-hadoop:latest 37.1s done
#10 DONE 93.5s
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451>
```



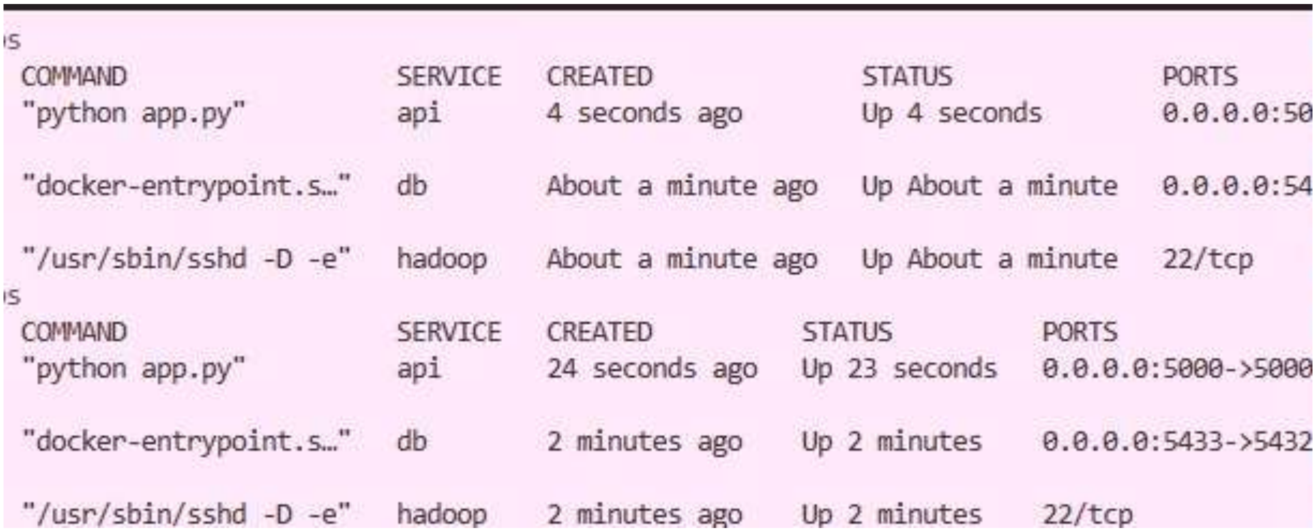
```
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451> docker run --rm examen-hadoop java -version
openjdk version "17.0.16" 2025-07-15
OpenJDK Runtime Environment (build 17.0.16+8-alpine-r0)
OpenJDK 64-Bit Server VM (build 17.0.16+8-alpine-r0, mixed mode, sharing)
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451>
```



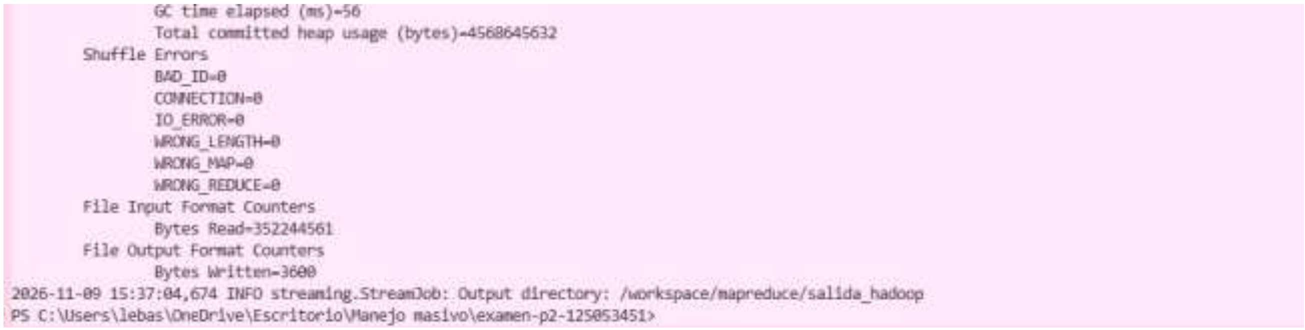
```
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451> docker run --rm examen-hadoop java -version
openjdk version "17.0.16" 2025-07-15
OpenJDK Runtime Environment (build 17.0.16+8-alpine-r0)
OpenJDK 64-Bit Server VM (build 17.0.16+8-alpine-r0, mixed mode, sharing)
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451>
```



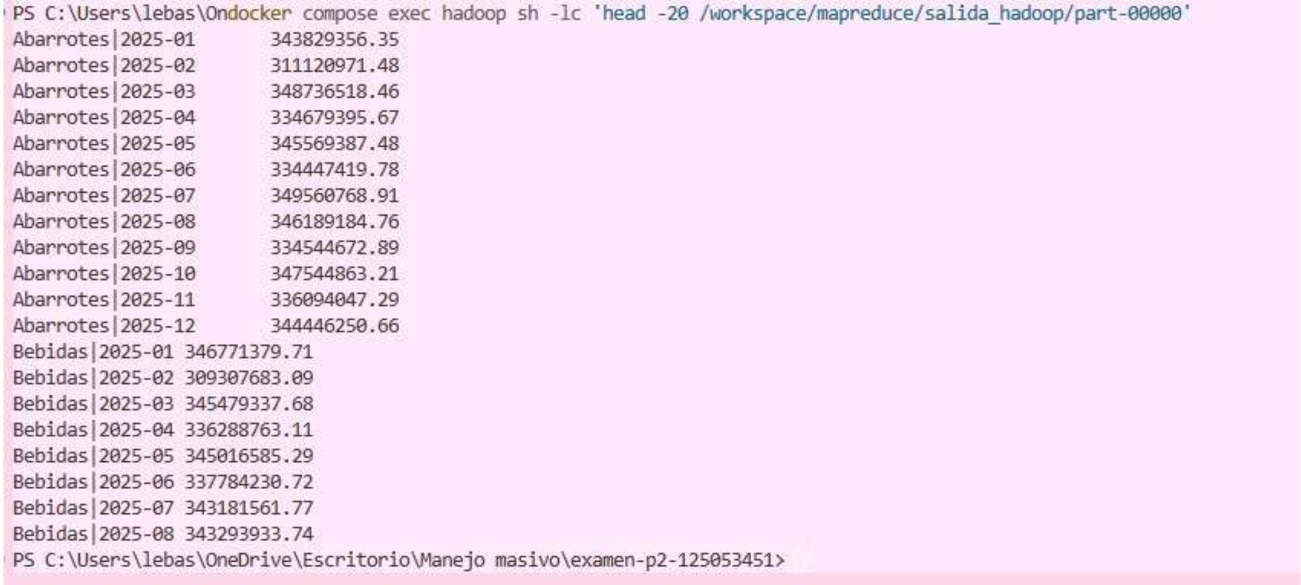
```
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451> docker compose ps
NAME            IMAGE
examen-api      examen-p2-125053451-api
00->5000/tcp, [::]:5000->5000/tcp
examen-db       postgres:16
33->5432/tcp, [::]:5433->5432/tcp
examen-hadoop   sha256:6d899dfbec18dc899a003fd5d92a9d294e754555cbcbd4ab20b7f0d94844a185
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451> docker compose ps
NAME            IMAGE
examen-api      examen-p2-125053451-api
/tcp, [::]:5000->5000/tcp
examen-db       postgres:16
/tcp, [::]:5433->5432/tcp
examen-hadoop   sha256:6d899dfbec18dc899a003fd5d92a9d294e754555cbcbd4ab20b7f0d94844a185
```



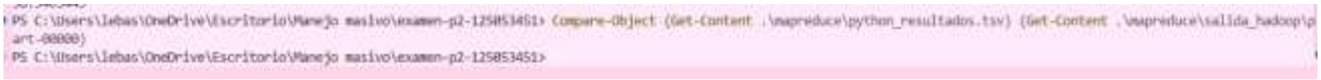
```
S
 COMMAND                  SERVICE   CREATED              STATUS               PORTS
 "python app.py"          api       4 seconds ago        Up 4 seconds         0.0.0.0:50

 "docker-entrypoint.s…"   db        About a minute ago   Up About a minute    0.0.0.0:54

 "/usr/sbin/sshd -D -e"   hadoop    About a minute ago   Up About a minute    22/tcp
S
 COMMAND                  SERVICE   CREATED          STATUS          PORTS
 "python app.py"          api       24 seconds ago   Up 23 seconds   0.0.0.0:5000->5000

 "docker-entrypoint.s…"   db        2 minutes ago    Up 2 minutes    0.0.0.0:5433->5432

 "/usr/sbin/sshd -D -e"   hadoop    2 minutes ago    Up 2 minutes    22/tcp
```



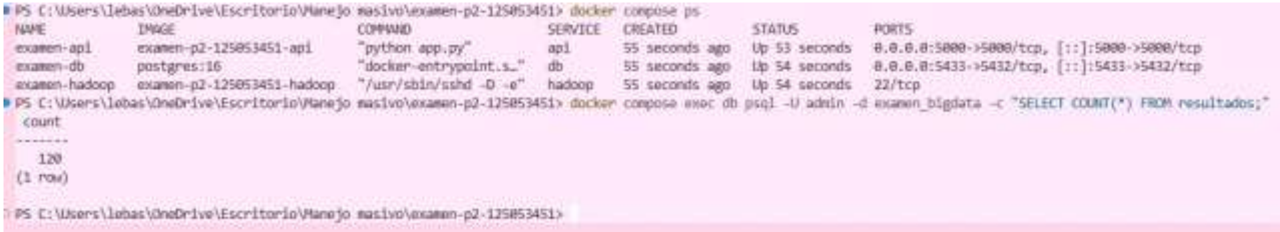
```
                GC time elapsed (ms)=56
                Total committed heap usage (bytes)=4568645632
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=352244561
        File Output Format Counters
                Bytes Written=3600
2026-11-09 15:37:04,674 INFO streaming.StreamJob: Output directory: /workspace/mapreduce/salida_hadoop
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451>
```



```
PS C:\Users\lebas\Ondocker compose exec hadoop sh -lc 'head -20 /workspace/mapreduce/salida_hadoop/part-00000'
Abarrotes|2025-01       343829356.35
Abarrotes|2025-02       311120971.48
Abarrotes|2025-03       348736518.46
Abarrotes|2025-04       334679395.67
Abarrotes|2025-05       345569387.48
Abarrotes|2025-06       334447419.78
Abarrotes|2025-07       349560768.91
Abarrotes|2025-08       346189184.76
Abarrotes|2025-09       334544672.89
Abarrotes|2025-10       347544863.21
Abarrotes|2025-11       336094047.29
Abarrotes|2025-12       344446250.66
Bebidas|2025-01 346771379.71
Bebidas|2025-02 309307683.09
Bebidas|2025-03 345479337.68
Bebidas|2025-04 336288763.11
Bebidas|2025-05 345016585.29
Bebidas|2025-06 337784230.72
Bebidas|2025-07 343181561.77
Bebidas|2025-08 343293933.74
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451>
```



```
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451> Compare-Object (Get-Content .\mapreduce\python_resultados.tsv) (Get-Content .\mapreduce\salida_hadoop\p
art-00000)
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451>
```



```
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451> docker compose ps
NAME             IMAGE                          COMMAND                  SERVICE   CREATED          STATUS          PORTS
examen-api       examen-p2-125053451-api        "python app.py"          api       55 seconds ago   Up 53 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
examen-db        postgres:16                    "docker-entrypoint.s…"   db        55 seconds ago   Up 54 seconds   0.0.0.0:5433->5432/tcp, [::]:5433->5432/tcp
examen-hadoop    examen-p2-125053451-hadoop     "/usr/sbin/sshd -D -e"   hadoop    55 seconds ago   Up 54 seconds   22/tcp
PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451> docker compose exec db psql -U admin -d examen_bigdata -c "SELECT COUNT(*) FROM resultados;"
 count
-------
   120
(1 row)

PS C:\Users\lebas\OneDrive\Escritorio\Manejo masivo\examen-p2-125053451>
```